

# Melanie Sclar

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## Education

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**University of Washington** 2026

PhD in Computer Science

*Advisors: Yejin Choi, Yulia Tsvetkov*

Dissertation: *Structure-Guided Approaches for Robust Language Model Reasoning*. GPA: 3.98/4.0.

**Universidad de Buenos Aires** 2017

Licenciatura en Ciencias de la Computación (6-year integrated Master's and Bachelor's degree)

Dissertation: *Analysis and Prediction of Human Visual Search*. GPA: 9.42/10 (equiv. 3.96/4.0).

## Work Experience

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**Research Scientist** Sept 2023–Sept 2025; May 2026–present

*Meta Superintelligence Labs (Fundamental AI Research, FAIR)*

(2026–; Research Scientist) LLM pretraining research team, focusing on novel losses and distillation techniques.  
(2023–2025; Visiting Researcher) Developed robust evaluation and synthetic data generation methods for theory of mind in LLMs (ICLR 2025) and interpretable personalized preference models (COLM 2025 Oral).

**Research Intern** Apr–Aug 2021

*Carnegie Mellon University / Advisors: Yonatan Bisk, Graham Neubig*

Designed first fully-symmetric multi-agent environment requiring theory of mind (ToM). Showed multi-agent RL models benefit from explicit ToM modeling but still achieve <50% of simple heuristics' performance (ICML'22).

**Lead Machine Learning Engineer** Nov 2019–Apr 2021

*ASAPP*

Built context-aware message suggestion system to speed up customer support interactions for Fortune 500 companies. Designed extensible architecture combining neural encoders with tree-based ranking, allowing rapid addition of new customer-vetted responses without model retraining (+15% CTR in the core company product).

**Lead Machine Learning Engineer** Jan 2018–Oct 2019

*BrightSector Algorithms*

Led 5-person team researching and deploying NLP algorithms at scale for Mercado Libre (Latin America's largest e-commerce platform). Developed named entity recognition models for product attribute extraction, classification systems for item categorization, and clustering methods to consolidate 200M+ listings.

**Software Engineering Intern** Jan–Apr 2016, Jan–Apr 2015

*Facebook Inc.*

(2016) Researched adaptive video stabilization to optimize compute while preserving video quality (2 patents).  
(2015) Built recommendation algorithms for Feed content using sentiment analysis and trending topic detection.

## Selected Honors & Awards *for research-related awards, see Publications*

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### Mathematics Olympiads

**National Champion**, Argentine Mathematical Olympiad (2011)

**Bronze Medal**, Ibero-American Mathematical Olympiad (2011)

**Silver Medal**, South American Mathematical Olympiad (2010)

Second place, Inter-University Argentine Mathematical Competition (CIMA) (2016)

Bronze Medal, Ibero-American Youth Mathematics Competition (2008)

### Computer Science and Machine Learning Olympiads

**Latin America Champion, ICPC World Finals** (2015) – International Collegiate Programming Competition  
**Champion, South America South Regional ICPC** (2013, 2014); 3rd place (2012)

**National Champion, Argentine ICPC Competition** (2014); 2nd place (2012); 3rd place (2013)

First (2017) and second place (2018), Universidad de Buenos Aires Computing School ML nationwide competition

## Selected Publications

*\*denotes equal contribution*

### Robustness, Model Behavior Quantification and Understanding

**Sclar, M.**, Choi, Y., Tsvetkov, Y., Suhr, A. (2024). Quantifying Language Models' Sensitivity to Spurious Features in Prompt Design or: How I learned to start worrying about prompt formatting. **ICLR 2024 (700+ cits)**.

**Sclar, M.\***, Dziri, N.\*, Lu, X.\*, et al. (2023). Faith and Fate: Limits of Transformers on Compositionality. **NeurIPS 2023, Spotlight (700+ citations)**.

Lu, X., **Sclar, M.**, [...], Choi, Y. (2025). AI as Humanity's Salieri: Quantifying Linguistic Creativity of Language Models via Systematic Attribution of Machine Text against Web Text. **ICLR 2025, Oral**.

► **Recipient of 2026 Open Philanthropy Grants** for (1) detecting benchmark gaming and data contamination (\$210,000) and (2) detecting interpretable predictive features of prompt quality in LLM evaluations (\$145,000).

### Theory of Mind Reasoning, Personalization, Modeling Human Behavior

**Sclar, M.**, Kumar, S., West, P., Suhr, A., Choi, Y., Tsvetkov, Y. (2023). Minding Language Models' (Lack of) Theory of Mind: A Plug-and-Play Multi-Character Belief Tracker. **ACL 2023, Outstanding Paper Award**. Awarded **Amazon Research Gift Grant (\$125,000)** based on this research to continue this line of work.

Li, SS., **Sclar, M.**, et al., PrefPalette: Personalized Preference Modeling with Latent Attributes. (2025). **COLM 2025, Oral**.

**Sclar, M.**, Yu, J., Fazel-Zarandi, M., Tsvetkov, Y., Bisk, Y., Choi, Y., Celikyilmaz, A. (2025). Explore Theory of Mind: Program-guided adversarial data generation for theory of mind reasoning. ICLR 2025.

**Sclar, M.**, Neubig, G., Bisk, Y. (2022). Symmetric Machine Theory of Mind. ICML 2022.

**Sclar, M.\***, Bujia, G.\*, Vita, S., Solovey, G., Kamienkowski, J.E. (2020). Modeling human visual search: A combined bayesian searcher and saliency map approach for eye movement guidance in natural scenes. SVRHM NeurIPS Workshop 2020, Oral presentation; **NVIDIA Diversity in AI Best Paper Award**.

## Patents

Wolf, W.A., **Sclar, M.**, et al. (2024). Processing clusters with mathematical models for message suggestion. U.S. Patent 11,985,102.

**Sclar, M.**, et al. (2020). Neural network to optimize video stabilization parameters. U.S. Patent 10,582,211.

**Sclar, M.**, et al. (2019). Foreground detection for video stabilization. U.S. Patent 10,506,248.

## Selected Service, Outreach & Teaching

**Teaching:** Head Teaching Assistant, University of Washington (two quarters); Teaching Assistant, Universidad de Buenos Aires (9 semesters; 2014–2017, 2019); Mathematics Olympiad Coach (2012–2016), plus others below.

**Organization:** Bridging Language, Agent, and World Models for Reasoning and Planning Workshop (NeurIPS 2025); South America Topcoder Open Regionals (2019); 20+ programming and mathematics competitions (2012–2018); Organizer & instructor at competitive programming training camps (2014–2018; Brazil & Argentina)

**Outreach:** Weekly programming instructor for underrepresented students in partnership with National University of San Martin, Argentina (2016–2018); Invited speaker at FemIT 2020, conference for Latin American women in tech (1500+ attendees); University of Buenos Aires' annual Computer Science Week Invited Speaker (2017–2020)

**Jury Member:** Argentine Informatics Olympiad (2018–present); Argentine Mathematics Olympiad (2020)

**Invited Speaker:** Visions of Language Modeling Workshop @ COLM (2025); Oracle ML (2024), CMU (2024), Universidad Torcuato di Tella (2024), among others.

## Skills

**Programming:** Python (advanced); C++, Java, MATLAB, SQL (intermediate); R, Bash, PHP/Hack (basic)

**Languages:** Spanish (native), English (fluent, TOEFL 119/120), Portuguese (advanced), French (intermediate)